# Air Quality Knowledge as a Driver of Computing Competences: An International Project-Based Learning Project

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**Abstract:** This paper presents key recommendations developed based on the analysis of an international didactic project conducted jointly by two universities. The initiative aimed to enhance students' competencies in data processing systems, software engineering, and the applications of computer science in environmental engineering. Utilizing real-world air quality data and modern technologies (Node-RED, RabbitMQ, Docker), the project provided students with hands-on experience in near-production environments. An analysis of the course outcomes and performance led to the formulation of targeted recommendations across three core pillars: Software Engineering: Emphasizes deeper integration of the "learning by doing" approach alongside formal engineering practices (documentation, automated testing, code quality control) to effectively balance rapid prototyping with professional industry standards. Environmental Application Design: Highlights the necessity of embedding the environmental context as the foundation of the design process, recommending the development of user-engaging features (gamification, mobile sensor data) to foster citizen science. Inter-University Cooperation: Advocates for a hybrid teaching model to maximize interaction, alongside the standardization of evaluation criteria and collaborative tools (e.g., Git) to better prepare students for a global professional environment. These guidelines offer a strategic framework for optimizing future educational initiatives that bridge theory, practice, and international academic synergy.

**Keywords:** Air quality, computing competence, system development, Software Engineering, Environmental Application Design.

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# 1 Introduction

In the era of progressive climate change and its compounding negative impacts, life in urban agglomerations is becoming increasingly challenging. Urban populations, alongside local flora and fauna, are highly vulnerable to the amplified frequency of extreme weather events and severe thermal stress. Anthropogenic climate models predict that the global average maximum temperature in European, South American, and African cities will escalate by 2 to 8°C over the coming decades [Cl26]. This systemic temperature increase, combined with shifting precipitation regimes, triggers a dangerous synergy with localized air pollution, drastically reducing the environmental carrying capacity of modern cities [Wi16].

The World Health Organization (WHO) classifies ambient air pollution as one of the most critical environmental hazards to human health. Prolonged exposure to elevated concentrations of atmospheric pollutants induces severe systemic health failures, predominantly manifesting as chronic respiratory diseases, acute asthma exacerbations, and ischemic strokes. Highlighting the scale of this crisis, the European Environment Agency (EEA) attributes substantial premature mortality rates directly to poor air quality, citing annual losses of 253,000 lives due to fine particulate matter ($PM_{2.5}$), 52,000 due to nitrogen dioxide ($NO_2$), and 22,000 due to ground-level ozone ($O_3$) (klimaoslo, 2025).

This crisis is heavily compounded by rapid global demographic shifts. Urbanization acts as a major driver of local microclimatic modification, often intensifying the Urban Heat Island (UHI) effect. Szałata et al. (2024) identify expanding urban structures as complex, concentrated sources of diverse pollutants, ranging from vehicular exhaust fumes and industrial emissions to heavy construction dust.

Addressing these interconnected challenges requires a fundamental realignment of municipal policies toward sustainable development goals [Gi22]. Derlukiewicz et al. (2023) argue that the most effective response to these urban vulnerabilities is the systemic implementation of low-carbon solutions and the structural transition toward "low-carbon city" frameworks. Crucially, as Fu et al. (2010) emphasize[FLW10], low-carbon urban planning is a multi-objective optimization process. It does not merely seek the reduction of greenhouse gas emissions, but simultaneously aims to secure decoupling-based economic growth and elevate the socioeconomic quality of life for citizens. According to contemporary frameworks, low-carbon initiatives represent an intersection of advanced technologies, mitigation strategies, and green practices designed to capture or reduce atmospheric carbon dioxide ($CO_2$) while fostering long-term urban resilience (klima.praha, 2025). Consequently, modern municipalities are increasingly forced to re-engineer their infrastructure—ranging from energy grids to real-time environmental monitoring networks—to mitigate emissions and safeguard public health.

Armed with knowledge of air quality and its impact on health, the authors of this publication attempted to develop an application solution aimed at implementing an app that would enable navigation around the Campus and its surrounding areas in Gdańsk and Minden, in areas with the best air quality. The work aimed to monitor the most optimal cycling routes and implement them into the operating system.

The authors of the publication, together with students, identified the most frequently used access roads to the Campus to verify them in terms of air quality.

The jointly developed application for air quality assessment, designed to integrate with publicly available applications, determines optimal access routes based on the environmental parameter of air quality. According to the authors, who represent an interdisciplinary perspective, this is a novelty that can be confidently implemented into publicly available solutions in the future, constituting a new area of research activity.

This publication presents key project assumptions, highlighting the potential of AI tools for environmental protection and building useful applications. It also outlines the research methodology and expands the chapter on Integration of Web Engineering and Environmental Education. Importantly, it presents recommendations for implementing the developed solutions and key conclusions drawn from the international project.

## 2 Student education model

Developing internationally coordinated, interdisciplinary student projects bridging hardware, server infrastructure, and artificial intelligence (AI) requires a multifaceted pedagogical approach. Establishing this framework relies on literature covering modern Internet of Things (IoT) pedagogy, distributed project management, AI-assisted coding, and machine learning (ML) for environmental data analysis [GA23], [Ka23].

IoT pedagogy has shifted from lectures to active, Project-Based Learning (PBL) and STEM Maker education. Integrating iterative engineering design—proposing questions, planning, implementing, and evaluating—bridges theoretical knowledge with physical outputs [Ch20]. Building tangible environmental monitors connects students directly with data collection, making abstract computational concepts accessible and boosting engagement [SÇS25].

Managing complex, multi-university projects requires robust technological and team management frameworks. Distributed IoT systems provide real-time insights into technical bottlenecks and student performance, allowing managers to adapt teaching methodologies and align distributed teams [GA23].

Simultaneously, Large Language Models (LLMs) and AI coding assistants (e.g., GitHub Copilot, ChatGPT) have transformed software engineering. In education, AI tools enhance programming efficiency, though students often struggle with over-reliance and evaluating code correctness [Ni24]. In professional settings, AI productivity relies on suggestion quality rather than time-to-completion, using telemetry models to suppress low-quality suggestions and minimize cognitive fatigue [Mo24]. However, commercial deployment still requires human oversight to meet production standards like accessibility [Mo25].

For environmental applications, low-cost IoT sensors face hardware noise and calibration drift. Server-side ML algorithms (e.g., Random Forest, XGBoost, Support Vector Regression) correct these errors, enabling real-time Air Quality Index (AQI) predictions and making student-built sensors viable for scalable monitoring [Ka23].

While existing literature addresses IoT pedagogy, project management, AI coding, and ML analysis separately, a research gap remains in synthesizing them into a cohesive higher-education curriculum. Combining tangible IoT development with advanced AI analysis provides a unified framework to teach web engineering, modern AI coding, and sustainable environmental applications within a globally coordinated initiative [Zi22].

To address this gap and evaluate the effectiveness of the proposed international didactic framework, this study formulates three primary research questions:

- RQ1 (Software Engineering & Pedagogy): How does integrating real-world environmental streaming data (e.g., Airly API, mobile sensors) into an international Project-Based Learning (PBL) framework affect students' acquisition of software engineering competencies compared to traditional simulated coding tasks?
- RQ2 (Didactics & AI/IoT Technologies): In what ways can flow-based data modeling tools (Node-RED) combined with message brokers (RabbitMQ) and AI-assisted workflows balance rapid prototyping with the enforcement of production-grade engineering standards in distributed student teams?
- RQ3 (Environmental Engagement & Citizen Science): To what extent does embedding domain-specific air quality knowledge (e.g., AQI, $PM_{2.5}/PM_{10}$, route exposure) and gamification mechanisms into web application design enhance both user engagement and student environmental literacy?

## 3 Research methodology

In the winter semester of the 2025/2026 academic year, Bielefeld University of Applied Sciences and Arts and WSB Merito University in Gdańsk executed a joint educational project. Supervised by Prof. Dr. Ing. Grit Behrens and Prof. dr hab. inż. Cezary Orłowski, the initiative aimed to build practical competencies in distributed data processing systems using real-world air quality data.

Environmental measurements—including NO, CO, $NO_2$, $O_3$, $SO_2$, PM2.5, PM10, and $NH_3$—were sourced via the Airly API (**Fig. 1**). Working with authentic data provided students with a production-like operating environment.

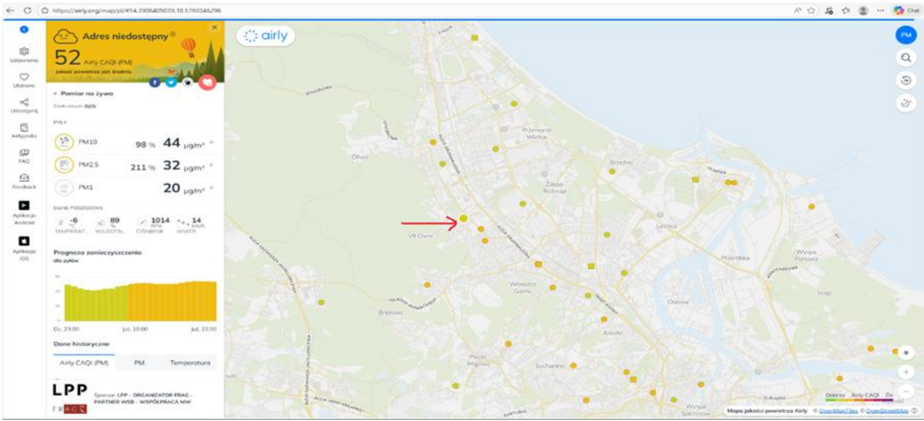


Fig. 1 Airly API: Our Data Source

Twelve students from WSB Merito University in Gdańsk implemented nine applications, each dedicated to a specific local air quality monitoring station. The course combined theoretical foundations delivered via Moodle with hands-on, experiential learning. Following an introduction to project objectives and distributed systems concepts, students entered the implementation phase using Node-RED for flow-based data pipeline modeling and RabbitMQ for asynchronous message queuing and routing.

The detailed data processing pipeline is presented in Figure 2 (Fig. 2 – Data Flow Steps), which illustrates the successive stages of data handling—from data acquisition via the API, through preprocessing and classification, to storage in structures enabling further analysis. This approach allowed students to gain a comprehensive understanding not only of data processing itself but also of asynchronous communication mechanisms and the scalability characteristics of distributed systems.

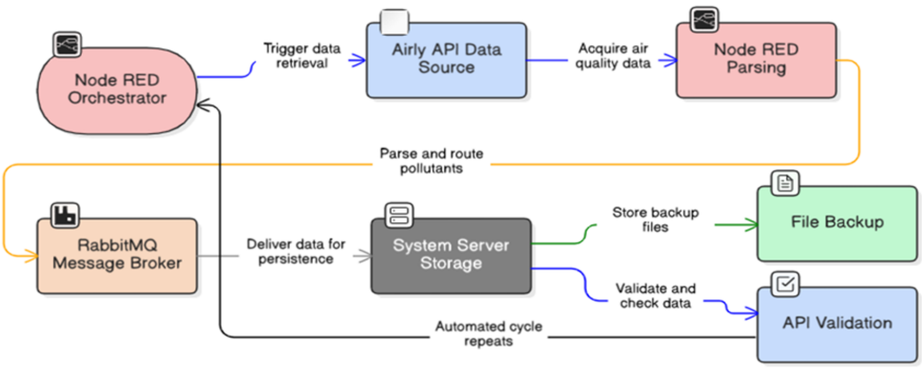

Fig. 2 Data Flow Steps

A key project aspect was preparing data for artificial intelligence systems, specifically Retrieval-Augmented Generation (RAG) paradigms. Processed RabbitMQ data were converted into structured textual documents to create knowledge bases for Large Language Models (LLMs). Students managed the complete solution lifecycle, including Docker containerization (docker-compose), Node-RED pipeline development, system monitoring, and technical presentations covering architecture and data flows.

The outcomes were presented in the form of technical presentations covering system architecture, component configuration, and data flow analysis. Monitoring mechanisms and potential use cases of the collected data are depicted in Figure 3 (Fig. 3 – Monitoring and Using the Data Collected), which highlights both technical and application-oriented perspectives.


Fig. 3 – Monitoring and Using the Data Collected

The project yielded nine functional applications and several key pedagogical insights:

- **Tooling Synergy:** Flow-based tools (Node-RED) combined with message brokers (RabbitMQ) simplified distributed system design and supported rapid prototyping ("vibe coding"), particularly for automated container configuration (e.g., docker-compose).
- **Online Limitations:** Fully remote instruction restricted communication and teamwork development, indicating that hybrid or in-person formats are better suited for collaborative skills.
- **Engineering Standards:** While individual projects accelerated implementation, they lacked formal software lifecycle rigor. Future iterations should incorporate testing, version control, and documentation

In conclusion, the project represents a valuable example of integrating academic education with practical applications of modern data processing technologies and artificial intelligence. The incorporation of real-world data sources, industry-relevant tools, and international collaboration significantly enhanced the quality of the learning process and contributed to better preparing students for the challenges of the contemporary labor market.

## 4 Integration of Web Engineering and Environmental Education

During the 2025/2026 winter semester at Hochschule Bielefeld (Campus Minden), the Web Engineering course applied a Project-Based Learning (PBL) approach connecting software engineering with healthy air initiatives. Students developed web-based components for the SmartMonitoring Airquality system, addressing urban air quality data collection, visualization, and analysis.

The methodology combined theoretical instruction, agile workflows, and functional prototyping. Following an international exchange lecture by Dr. Łukasz Szałata (Wrocław University of Science and Technology) on air quality metrics (PM2.5, PM10, $NO_x$, $O_3$, temperature), students translated environmental domain knowledge into technical requirements, database models, APIs, and web applications.

The course utilized real-world stationary data from Gdansk (Airly API) alongside stationary and mobile bicycle-based sensor data from Minden. Incorporating cycling added a citizen-science dimension, allowing students to analyze spatiotemporal pollution patterns, compare routes, and identify high exposure areas. Development was divided into specialized strands, including a Researcher Interface featuring map visualizations, route management, sensor overviews, statistical summaries, and configurable plots for decision-makers.

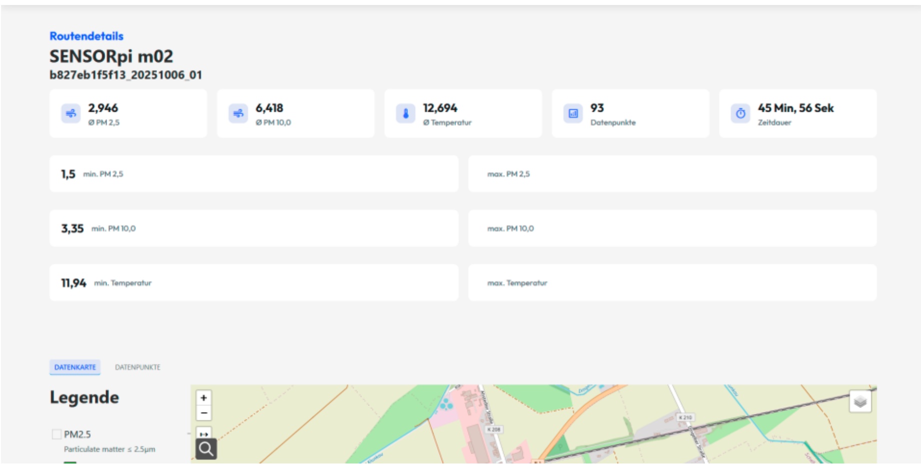

Fig. 4 - Web page for route details of mobile bicycle sensor with values for mean PM2.5 of 2.9, mean PM10 of 6.418, mean temperature, number of measurements along the route, and duration of the route.

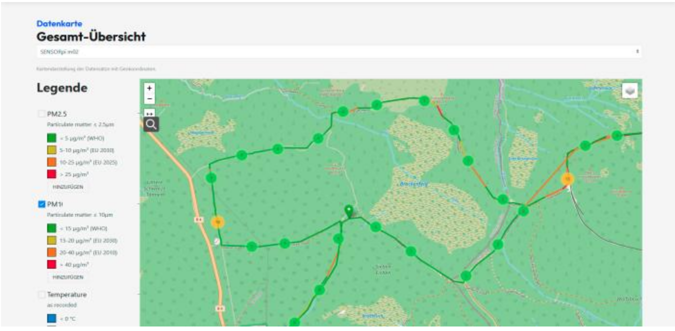


Fig.5 - Visualization of a bicycle route in a green area around mountain Brocken mountain in Germany in students' web application

Another group developed a Progressive Web Application (PWA) with push notifications and gamification mechanisms—including achievements, streaks, and leaderboards—to incentivize regular user participation in environmental data collection.


Fig. 6. Achievements designed by the student team for motivation for air-quality measurements and used in the gamified web application prototype.

From a software engineering perspective, students applied full-lifecycle practices—including requirements analysis, architecture and database modeling, REST APIs, Git-based workflows, issue tracking, and iterative development. The technological stack comprised the SmartData API, Smart Web Application Components, PostgreSQL/PostGIS, Payara, Swagger/OpenAPI, and PWA capabilities (Service Workers, Push API).

This approach integrated three core learning dimensions:

- Technical Mastery: Hands-on competencies in web engineering and distributed systems.
- Domain Context: Deeper understanding of urban air quality issues.
- Environmental Literacy: Leveraging informatics to transform raw data into visible, actionable insights for sustainability and healthier mobility choices.

## 5 Recommendations

Based on the evaluation of this joint international initiative, three key sets of recommendations are proposed to optimize future educational programs integrating software engineering, distributed systems, and environmental analytics:

1. Software Engineering Didactics
   a) Balance Prototyping with Rigor: While tools like Node-RED, RabbitMQ, and Docker accelerate architectural understanding through "learning by doing", future iterations must strictly integrate full software lifecycle practices—specifically automated testing, systematic documentation, version control, and code quality assurance.
   b) Near-Production Environments: Practical projects should continuously rely on authentic data streams to bridge the gap between academic theory and industrial software deployment.
2. Application Design in Environmental Engineering
   a) Domain-Centric Design: Environmental context must serve as the structural foundation of the system (dictating requirements, data models, and UI) rather than a superficial dataset.
   b) Citizen Science & Engagement: Integrating gamification, push notifications, and mobile sensor data (e.g., bicycle-based measurements) enhances spatiotemporal analysis, encourages user participation, and drives interdisciplinary learning.
3. Multi-University Collaborative Frameworks
   a) Hybrid Instruction Model: Purely remote delivery hinders teamwork and communication. A hybrid approach—combining online collaboration with targeted in-person sessions—is essential for building soft competencies.
   b) Standardized Infrastructure: Joint courses require unified work environments (e.g., shared Git platforms, project management tools), clear role allocation, and aligned evaluation standards to facilitate effective cross-institutional knowledge transfer.

## 6 Discussion and conclusions

The outcomes of this joint international project provide direct insights into the three formulated research questions. Regarding RQ1, the exposure to authentic, noisy environmental data streams significantly elevated students' engagement and forced them to address real-world system edge cases, which simulated coding environments fail to replicate. Addressing RQ2, while the combination of Node-RED, RabbitMQ, and Docker facilitated rapid architectural prototyping ("vibe coding"), it highlighted the vital necessity of enforcing strict software lifecycle practices—such as automated testing and continuous documentation—to maintain production quality. Finally, concerning RQ3, placing air quality domain knowledge at the core of the application design (e.g., eco-routing, gamification, and mobile sensor tracking) successfully fostered citizen science principles and elevated students' broader environmental literacy.

The authors clearly emphasize that international cooperation forms the cornerstone of effective research and educational initiatives. By combining the expertise of academics and students across universities around a shared objective—the development and implementation of the "Air" project—this initiative established an innovative international framework for data processing and forecasting systems.

The core strength of the project lies in placing the environmental context at the very center of the engineering lifecycle: air quality parameters directly shaped the requirements analysis, database models, and user interfaces. This domain-driven approach allowed students to develop a highly functional tool for data processing and air-quality-aware route optimization, while substantially strengthening their practical IT competencies.

To ensure the success and sustainability of future cross-border teaching initiatives, the authors highlight several structural insights:

- Incentive Frameworks for Online Collaboration: Managing remote, multi-institutional teams requires significant effort. To boost engagement, academic curricula and examination regulations should be adapted to award extra ECTS credits to participating students and credit additional teaching hours to instructors.
- Coherent Project Structure: Joint university projects demand clearly defined goals, clear role distribution, and intensive student-instructor interaction to overcome the inherent challenges of remote delivery.
- Multi-Format Educational Model: A multi-format teaching approach that systematically reinforces formal software engineering practices across consecutive semesters provides high educational value, preparing students for real-world labor market challenges.

Ultimately, this project demonstrates how merging computer science education with environmental awareness creates a valuable, replicable model for international academic collaboration.

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